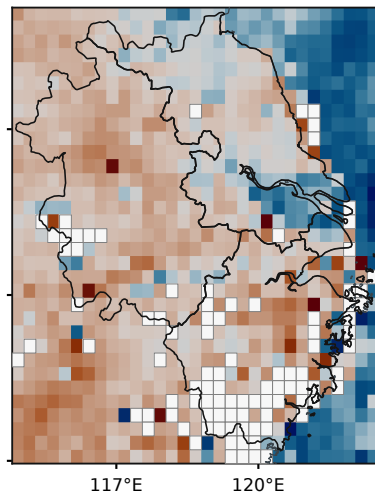
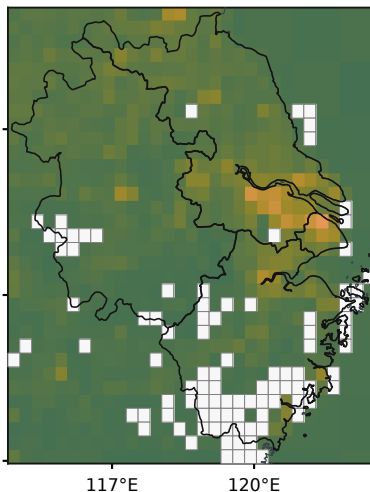
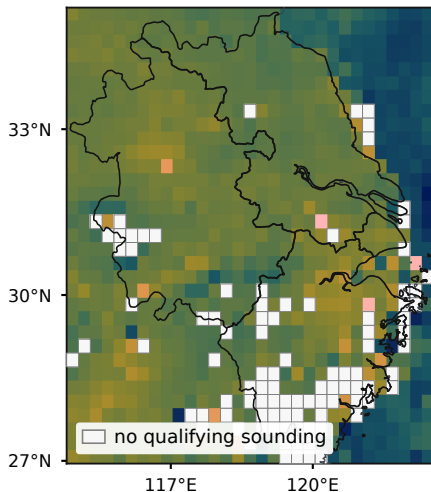


**(a) observed methane,
2018 annual mean**

**(b) the field impervious
fraction produces**

**(c) observed minus
that field**



1840 1860 1880 1900 1920 1940

-45.0 -22.5 0.0 22.5 45.0

Mean XCH_4 , bias corrected (ppb): one scale over (a) and (b)

Observed minus model field (ppb)

(b) is the field one land-cover covariate produces, held out -- fitted on the training rows of a spatial block and predicted onto rows the fit never saw -- and is not a prediction of methane; its held-out R squared is 0.085. (a) and (b) share one unclipped scale, 1840 to 1947 ppb: the observed field spans 106.1 ppb and the model field 42.7. (c) is symmetric about zero at plus or minus 45 ppb with arrow caps; 4 of 926 cells run past an end. 97 cells received no qualifying sounding and are absent in all three panels: an unobserved cell has no residual, not a residual of zero. The two limbs of (c) are matched in luminance so that equal errors of either sign read as equal, so the sign is carried by hue alone and a greyscale print of (c) shows the size of an error and not its direction. Moran's I of the residual 0.646, against 0.709 for the observed field, under queen contiguity among observed cells, row standardised, self excluded, no distance decay; pseudo $p \leq 0.001$ over 999 permutations, 1 cell with no observed queen neighbour dropped. See ERRATA.md 7.1 and 1.1.